METAL SOLUTIONS

EOS StainlessSteel 254

Material Data Sheet

EOS STAINLESSSTEEL 254

EOS StainlessSteel 254 is an austenitic stainless steel for extreme conditions. The high chromium, molybdenum and nitrogen alloying give excellent corrosion resistance in many difficult environments. The general pitting resistance equivalent PREN for 254 is 43 (PREN = %Cr + 3.3 X %Mo + 16 X %N).

MAIN CHARACTERISTICS

- Excellent resistance to uniform, pitting and crevice corrosion
- High resistance to stress corrosion cracking
- Higher strength than conventional austenitic stainless steels

TYPICAL APPLICATIONS

- Chlorinated seawater handling equipment
- Pulp and paper manufacturing devices
- Chemical handling equipment

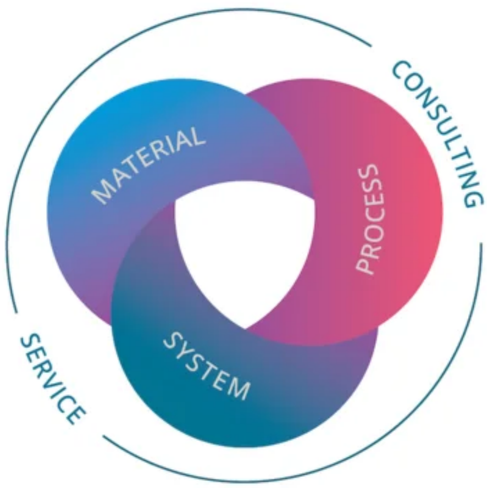
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

EOS StainlessSteel 254 powder material is in accordance with DIN EN 10088-3, EN 1.4547

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Cr	19.5	20.5
Ni	17.5	18.5
Mo	6	7
Cu	0.5	1
N	0.18	0.25

Powder Particle Size

Generic Particle Size Distribution	20 - 65 µm

HEAT TREATMENT

Description

Heat treatment procedure

Steps

Optional solution annealing: At 1180 °C for 2 h after parts have fully heated through, water quenching Typical dimensional change after heat treatment: 0.06 %

PROCESS DATA SHEET

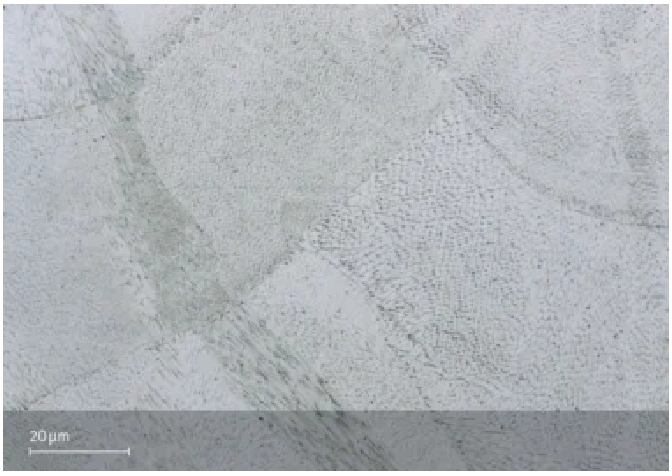
EOS StainlessSteel 254 for EOS M 290 | 40 µm

EOS M 290 - 40 µm - TRL 3

System Setup	EOS M 290
EOS Material set	254_040_CoreM291_100
Software Requirements	EOSPRINT 2.8 or newer EOSYSTEM 5.20 or newer
Recoater Blade	HSS (High Speed Steel)
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	4.1 mm^3/s

Chemical and Physical Properties of Parts



Micrograph etched as manufactured Etchant: ASTM E407-07, etchant 12

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	0.01 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	40 µm	≥ 8.07 g/cm^3	-

Mechanical Properties

Mechanical Properties Heat Treated

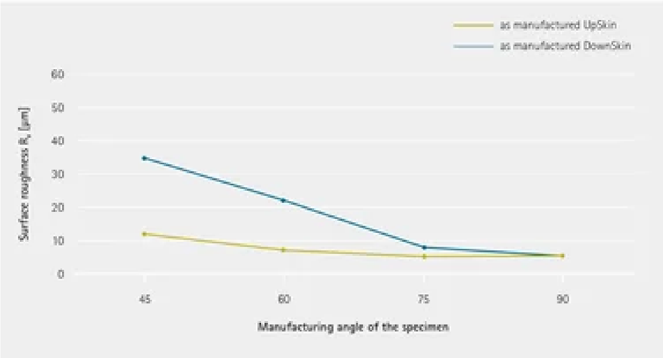
EN ISO 6892-1 Room Temperature | 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	360	660	48	-	-	-
Horizontal	360	700	43	-	-	-

Optional solution annealing
At 1 180 °C for 2 h after parts have fully heated through, water quenching
Typical dimensional change after heat treatment: 0.06 %

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature |	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	600	720	35	-	-	-
Horizontal	680	810	29	-	-	-

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
$14.8 \cdot 10^{-6}$/K	25 – 100 °C
$15.7 \cdot 10^{-6}$/K	25 – 200 °C
$16.3 \cdot 10^{-6}$/K	25 – 300 °C
$16.7 \cdot 10^{-6}$/K	25 – 400 °C

PROCESS DATA SHEET

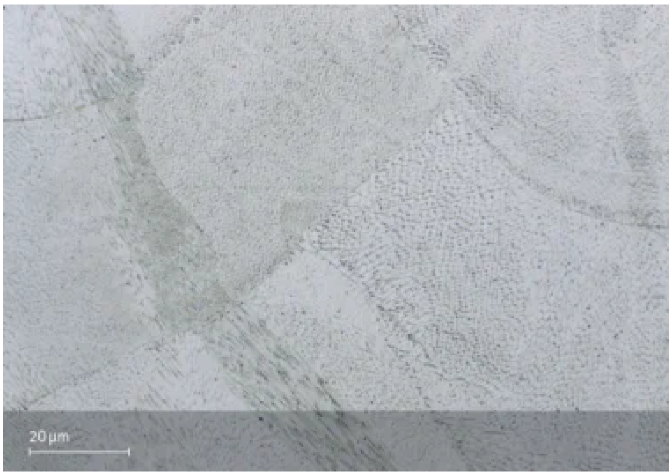
EOS StainlessSteel 254 for EOS M 290 | 60 µm

EOS M 290 - 60 µm - TRL 3

System Setup	EOS M 290
EOS Material set	254_060_CoreM291_100
Software Requirements	EOSPRINT 2.8 or newer EOSYSTEM 5.20 or newer
Recoater Blade	HSS (High Speed Steel)
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	60 µm
Volume Rate	6.1 mm^3/s

Chemical and Physical Properties of Parts



Micrograph etched as manufactured Etchant: ASTM E407-07, etchant
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Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	60 µm	0.02 %	-

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	60 µm	≥ 8.07 g/cm³	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature | 60 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	360	660	48	-	-	-
Horizontal	360	700	44	-	-	-

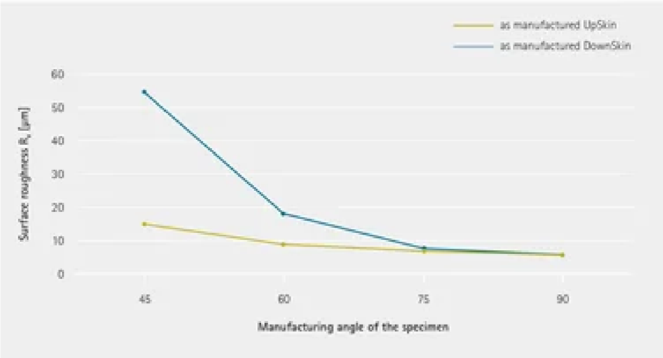
Optional solution annealing
At 1 180 °C for 2 h after parts have fully heated through, water quenching.

Typical dimensional change after heat treatment: 0.06 %

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature | 60 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	580	730	36	-	-	-
Horizontal	660	800	30	-	-	-

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
$14.8*10^{-6}$/K	25 – 100 °C
$15.7*10^{-6}$/K	25 – 200 °C
$16.3*10^{-6}$/K	25 – 300 °C
$16.7*10^{-6}$/K	25 – 400 °C

HEADQUARTERS

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